

Assessing Decision Reliability in Discrete-Event Simulation: A Quantum-Network Case Study

David Vesterlund¹

¹ *Vesterlund Ventures, WestQuant Open**
(Dated: 8 October 2026)

Simulation-based design decisions depend on stochastic variation, analyst choices, and model semantics. We present a workflow that separates these sources of uncertainty using a canonical experiment specification, independent seed pools, explicit intervention gains, and distinct measures of numerical, ranking, decision, and claim agreement. A quantum-network case study compares an author-developed simulator with a harmonized reference model; two framework adapters share the same reference logic and do not constitute independent model replications. Across 400 configurations, a reported two-pool experiment increases label agreement from 71.5% with a single seed to 95.8% with 20-seed averaging. The full protocol and simple averaging achieve the same agreement, so the improvement is attributable to replication rather than a uniquely superior labeling rule. Separate analyses show sensitivity to intervention strength and performance definitions. In a finite-grid memory analysis, 50.8% of 356 matched interior cases differ by at least a factor of four between model variants, with a configuration-clustered 95% interval of 37.4–63.9%. This empirical comparison is conditional on its metric, retained configurations, and shared implementation assumptions. The methodological contribution is an auditable separation of within-model repeatability, between-model sensitivity, and metric incompatibility. We provide reporting requirements and a staged verification plan; full native-protocol and hardware validation remain future work.

I. INTRODUCTION

Quantum-network design requires deciding which physical resource to improve first: memory capacity, generation rate, retrieval efficiency, coherence time, or routing. A natural approach is *intervention-based bottleneck identification*: improve each resource by a fixed amount, measure the marginal performance gain ΔG_j , and label the bottleneck as the resource with the largest ΔG_j . This approach is intuitive, model-agnostic, and produces a single actionable label.

But how robust is that label? A reviewer of any such analysis should ask: would the label change with a different random seed? With a larger or smaller intervention? With a different performance metric or fidelity threshold? With a different simulator?

A growing number of discrete-event simulators—NetSquid [1], SeQUeNCe [2], QuISP [3], SimQN [4]—offer different modeling choices for the same physical phenomena. When different models are applied to the same problem, they may produce different numerical predictions. This is not a bug; it is a consequence of the fact that simulation is an abstraction. The question is not whether simulators disagree—they inevitably do—but *which types of scientific conclusions survive a change of simulation model, and which do not*.

A. Simulator Output as a Function of Assumptions, Semantics, and Implementation

A simulator’s output depends on three layers:

1. **Model assumptions:** The physical and mathematical models (noise model, generation model, swapping model). These are typically well-documented in publications.
2. **Protocol semantics:** The operational definitions of what constitutes a “completed request,” when a latency clock starts, how waiting and queueing work, when memory slots are locked and released, whether generation is sequential or concurrent, and how deadlines and censoring are handled. These are often *implicit* and may differ between simulators without being documented.
3. **Implementation:** The software engineering (event scheduling, RNG, numerical precision, data structures). These affect reproducibility but not (in principle) scientific conclusions.

This distinction is critical. Two simulators that share the same model assumptions but differ in protocol semantics may produce different scientific conclusions. Two simulators that share both model assumptions and protocol semantics but differ in implementation should produce identical results (up to numerical precision). We therefore write simulator output as:

$$y = f(x, \text{model assumptions,} \quad (1) \\ \text{protocol semantics, implementation}).$$

and the trustworthiness of any conclusion depends on which of these factors it is sensitive to. This reframes the question from “which simulator is best?” to “what should a trustworthy quantum-network simulator guarantee for this conclusion to be reliable?”

* david@vesterlundventures.se; ORCID: 0009-0000-6455-1141

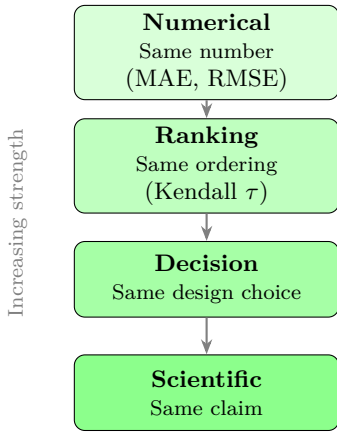


FIG. 1. Four types of cross-simulator agreement, ordered by strength. Numerical agreement is the weakest; scientific-conclusion agreement is the strongest. Two simulators can have high numerical agreement but low decision agreement (small differences near a threshold flip the decision). Each main result in this paper is reported on all four levels.

B. Four Types of Agreement

A central conceptual contribution is the distinction between four types of cross-simulator agreement, which are often conflated (Figure 1):

1. **Numerical agreement:** Two simulators produce the same number (or nearly the same number) for a given metric. Measured by MAE, RMSE, or relative error.
2. **Ranking agreement:** Two simulators produce the same ordering of configurations. Measured by Kendall τ or Spearman ρ .
3. **Decision agreement:** Two simulators lead to the same design decision (e.g., the same memory family is selected, the same topology is preferred). Measured by decision-changing disagreement rates.
4. **Scientific-conclusion agreement:** Two simulators support the same scientific claim (e.g., “memory family A outperforms B,” “throughput increases monotonically with memory,” “scalar collapse fails”). This is the strongest form of agreement and the one that matters for science.

These four types are not interchangeable. Two simulators can have low numerical agreement but high ranking agreement (all values are shifted by a constant factor, but the ordering is preserved). Two simulators can have high numerical agreement but low decision agreement (small differences near a threshold flip the decision). Throughout this paper, each main result is reported on all four levels where applicable.

C. Contributions

1. A unified fragility framework measuring three axes—noise, analyst choices, and model switching—on a shared Canonical Experiment Specification, with each result reported on four agreement levels.
2. A benchmark ladder (B0–B6) with the rule that each step must be passed before the next runs, replacing the ad-hoc CV0–CV7 ladder.
3. A semantic-dimensions table covering only the two systems actually run (WestQuant and SeQUeNCe), identifying which semantic choices are switchable and which are definition mismatches.
4. Empirical evidence that the dominant bottleneck label is fragile: 50.7% of configurations show at least one seed disagreement at 2000 requests; 63% of short-run changes are statistically tied.
5. A factorial design over semantic switches showing that retrieval-in-delivery explains 72% of A_{req} variance for AFC, but this measures A_{req} variance across configurations (including m and λ), not model disagreement. With all seven interventions, $\kappa = 0.055$ (retrieval on) and $\kappa = 0.000$ (retrieval matched); retrieval matching does not improve agreement.
6. A three-class taxonomy of conclusion reproducibility: structural (reproduced), model-sensitive (not reproduced), and definition-sensitive (non-comparable).
7. A decision-transfer experiment showing 0% transfer with retrieval on and 15% with retrieval matched; m_c disagreement (ratio $\geq 4\times$) is 85% in both conditions.
8. A Trust Stack checklist for reporting, with reviewer recommendations.

II. SETUP: COMMON EXPERIMENT SPECIFICATION AND VALIDATION LADDER

A. Canonical Experiment Specification

When two simulators are given “the same” scenario, they may still produce different results because the scenario was translated differently. Different RNG implementations generate different arrival traces. Different topology representations assign different node labels. This creates *false disagreement*—apparent simulator differences that are actually translation artifacts.

A Canonical Experiment Specification (CES) eliminates false disagreement by specifying every parameter before translation to any implementation. The CES defines:

- **Network:** Topology graph, node count, link distances.
- **Hardware:** Memory family, T_1 , T_2 , retrieval efficiency, readout fidelity.
- **Memory:** Capacity m , slot model, aging model.
- **Links:** Generation probability p_{gen} , attempt time, loss model.
- **Traffic:** Arrival process, arrival rate λ , source-destination pairs, deadline.
- **Protocol:** Swapping probability, purification, scheduling.
- **Routing:** Path selection algorithm.
- **Noise:** Depolarization, amplitude damping, readout errors.
- **Metrics:** Definitions of throughput, fidelity, latency, resource use.
- **Randomness:** Seeds, RNG algorithm, exogenous generation outside simulators.

The CES is simulator-neutral: it does not use any simulator’s parameter names or data structures. Each simulator provides a translator that maps CES parameters to its own configuration format. Exogenous randomness (topology, arrivals, source-destination pairs) is generated outside the simulators to eliminate false disagreement from different RNG implementations. Each scenario uses three independent seed streams: (i) a *graph seed* for topology generation, (ii) a *traffic seed* for request arrivals and source-destination pairs, and (iii) a **physics seed** for link generation outcomes, swap outcomes, and decoherence realizations.

B. Validation Ladder B0–B6

We propose a benchmark ladder for quantum-network simulation, with the rule that *each step must be passed before the next runs*. Table I summarizes the ladder and the validation criterion for each level.

The ladder is cumulative: a simulator that passes B4 has also passed B0–B3. The hardest level (B6) requires claim-level replication across model families. Hardware-calibrated validation against experimental data is a further level that remains future work; we do not have experimental data for contention-dominated scenarios.

B0 analytical verification gate. B0 tests single-link generation with parameterized p_{gen} , T_2 , and retrieval efficiency η . For a single link without contention and without retrieval loss, the analytical completion rate is $A_{\text{req}} = 1.0$. The harmonized model passes this gate because it treats retrieval as free ($\eta_r = 1$). WestQuant shows lower A_{req} (mean 0.65) because it counts retrieval

efficiency η_r in the delivery: a request is dropped if retrieval fails ($\eta_r < 1$). The analytical reference of 1.0 is correct only for a model that omits retrieval loss; a model that includes retrieval loss should give $A_{\text{req}} = \eta_r$, not 1.0. The B0 gate therefore rewards the model that omits the loss. This is a model-semantic effect (the `retrieval_in_delivery` switch, Sec. VB), not a physics error. WestQuant is retained because its retrieval semantics represent a deliberate modeling choice.

C. Simulator Frameworks

WestQuant is the native simulator. It models quantum networks with per-node memory capacity m , Bernoulli link generation, depolarizing noise, entanglement swapping, and concurrent request processing with queueing. When all memory slots at a node are occupied, incoming requests queue until a slot frees, incurring queueing delays that affect latency and memory aging.

SeQUeNCe and SimQN (Harmonized Mode) are used via harmonized adapters: their event engines are instantiated, but the physics logic is implemented in a shared harmonization layer using the same Bernoulli generation, depolarizing noise, and sequential blocking model. When memory is full, requests are rejected immediately rather than queued.

Event-trace audit. The SeQUeNCe and SimQN adapters produce bit-identical event traces across all tested scenarios (verified by SHA-256 hashing). The comparison is therefore effectively between one native model (WestQuant: concurrent generation + queueing) and one harmonized reference model (serial generation + immediate blocking), with $S_{\text{models}} = 2$ rather than 3. Treating SeQUeNCe and SimQN as two independent models would constitute pseudoreplication.

Native SeQUeNCe pilot. A native SeQUeNCe adapter was developed as a pilot, instantiating SeQUeNCe’s own `RouterNetTopo`, `QuantumRouter`, `BSMNode`, and `RequestApp` components. This adapter uses SeQUeNCe’s own components but does not execute native protocol events; it is therefore a partial validation, not a full native comparison. The pilot ran 3 scenarios $\times$ 3 seeds = 9 scenario-seed pairs. Native SeQUeNCe produces different A_{req} values from WestQuant in several cases (e.g., chain $N=3$ $m=2$: WQ 0.889 vs. native 0.667), confirming that the adapter produces different values from WestQuant. However, the pilot is too small for statistical comparison (wall time ~ 24 – 86 s per scenario vs. < 0.01 s for WestQuant). A larger independent-judge experiment (25 configurations $\times$ 10 seeds = 250 runs) is reported in Section VIII; it also uses SeQUeNCe components but does not execute native protocol events, so it remains a partial validation.

TABLE I. Benchmark ladder B0–B6. Each step must be passed before the next runs. B6 in this study is claim-level replication; hardware-calibrated validation (hardware-calibrated validation) remains future work.

Level	Name	Description	Validation Criterion	Scenarios
B0	Analytical primitives	Single-link generation, loss, decoherence	Match analytical rate formulas ($\epsilon_{\text{abs}} < 0.05$)	90
B1	Elementary entanglement	Two-node entanglement generation, memory families	Match analytical or experimental rates	180
B2	Three-node repeater	Single swap, finite memory	Match analytical swapping formulas	360
B3	Multi-hop chain	N -hop chain, path length scaling	Monotonicity, rate decay	180
B4	Finite memory + contention	Memory limit, queueing, multiple requests	Cross-simulator ranking agreement (τ)	200
B5	Heterogeneous topology + traffic	Multiple topologies, stochastic traffic	Structural conclusion agreement	60
B6	Claim-level replication	Full claim grid (C1–C5) across model families	Conclusion-level reproducibility	2,880

D. Performance Metrics

All implementations report three primary metrics, computed over the observation window $[t_{\text{warmup}}, T_{\text{obs}}]$ with warmup excluded:

$$A_{\text{req}} = \frac{N_{\text{completed}}}{N_{\text{arrived}}} \quad (2)$$

$$A_{\text{qual}} = \frac{N_{\text{completed}} \cap N_{\text{qualified}}}{N_{\text{arrived}}} \quad (3)$$

$$R_{\text{qual}} = \frac{N_{\text{completed}} \cap N_{\text{qualified}}}{T_{\text{obs}} - t_{\text{warmup}}} \quad (4)$$

where N_{arrived} is the number of requests that entered the system during the observation window, $N_{\text{completed}}$ are requests that reached the destination within the deadline $D = 10$ s, and $N_{\text{qualified}}$ are completed requests whose end-to-end fidelity $F \geq F_{\text{min}} = 0.5$. $T_{\text{obs}} = 200$ s is the observation horizon and $t_{\text{warmup}} = 20$ s is excluded. A_{req} and A_{qual} are fractions in $[0, 1]$; R_{qual} is a rate.

E. Intervention-Based Bottleneck Identification

Given a network configuration θ and a performance metric $G(\theta)$, we define the *intervention gain* for resource j as

$$\Delta G_j(\theta, \delta) = G(T_{j,\delta}(\theta)) - G(\theta), \quad (5)$$

where $T_{j,\delta}$ is an explicit intervention operator that improves resource j by a relative amount δ . The *soft bottleneck vector* is

$$p_j = \frac{\max(0, \Delta G_j)}{\sum_k \max(0, \Delta G_k)}, \quad (6)$$

and the *dominant bottleneck* is $\arg \max_j p_j$. The bottleneck entropy $H_B = -\sum_j p_j \log p_j$ measures how diffuse

the bottleneck is. If all $\Delta G_j \leq \epsilon$ ($\epsilon = 10^{-6}$), the configuration is labeled *degenerate*.

We study seven resources: **capacity** (memory slots), **generation** (success probability), **retrieval** (readout efficiency), **coherence** (decoherence rate), **routing** (policy), **deadline** (time limit), and **topology** (edge density).

F. Agreement Measures

We use four agreement measures, corresponding to the four types in Section IB:

- **Numerical:** MAE or RMSE against reference.
- **Ranking:** Kendall τ for rankings of the soft vector.
- **Decision:** Cohen’s κ for dominant labels; decision-changing disagreement rates.
- **Scientific conclusion:** classification as structural, model-sensitive, or definition-sensitive.

We also report Jensen–Shannon distance for the full soft vector distributions.

III. NOISE: STOCHASTIC FRAGILITY

A. Stratified Sample

We generate a stratified sample of 400 configurations covering 6 memory families (trapped ion, neutral atom, diamond electron, diamond nuclear, AFC, superconducting), 3 topologies (chain, star, RGG), 3 network sizes ($N \in \{10, 20, 50\}$), 3 traffic levels, and 3 budget levels. This yields $6 \times 3 \times 3 \times 3 \times 3 = 486$ strata, with 1–2 configurations per stratum. Each simulation run uses 2000 requests (Poisson arrivals), $T_{\text{obs}} = 200$ time units, no warmup, and depolarizing noise at medium level.

B. Four Procedures

We assess stochastic fragility with four procedures, each reported on all four agreement levels:

Procedure 1: Label stability. Using 5 seeds per configuration with common random numbers (same seed for both models), we measure label stability as the fraction of seeds agreeing with the modal dominant label. For A_{qual} , the mean stability was 0.686 (WestQuant) and 0.571 (Harmonized). The label changed for 50.7% of configurations across seeds (at least one seed disagreed); 28.2% showed a majority change (stability < 0.5).

Procedure 2: Confidence intervals and statistical ties. To assess whether the 79% “any change” figure at 100 requests reflects genuine instability or insufficient statistical evidence, we computed bootstrap confidence intervals (1000 resamples, 95% level) for the paired difference $\Delta G_{(1)} - \Delta G_{(2)}$ on a subset of 100 configurations with 20 seeds each. A configuration is *statistically tied* if the CI includes zero. Results: 91% showed any change, 58% were statistically tied, and **63% of “any change” configurations were tied**—the data are insufficient to distinguish the top two resources at this sample size. Only 34% showed non-tied label change (label changes AND CI excludes zero).

Procedure 3: Convergence control. We ran 30 configurations with $n_{\text{requests}} \in \{50, 100, 200, 500, 1000\}$ and $n_{\text{seeds}} \in \{5, 10, 20, 50\}$. At 20 seeds, label stability increased from 0.480 (50 requests) to 0.622 (1000 requests)—a 30% improvement. “Any change” decreased from 66.7% to 53.3%; “majority change” decreased from 36.7% to 20.0%. A full rerun at 2000 requests with 5 seeds confirms the trend: any-change drops to 50.7% and majority-change to 28.2%. The residual 50.7% any-change rate at 2000 requests suggests that while much of the short-run fragility is Monte Carlo noise, a genuine residual persists.

Procedure 4: Performance loss. For the 42 configurations (out of 100) that were not statistically tied, the mean relative loss from choosing the second-best resource instead of the best was 58.5% of the best gain. 62% of non-tied configurations lose more than 50% of the best gain if the second-best is chosen.

C. Agreement Levels for Noise

- **Numerical:** Mean label stability 0.686 (WQ) at 2000 requests; MAE in ΔG_j varies by resource.
- **Ranking:** Kendall τ across seeds is not directly applicable (same model, different seeds); instead, label stability serves as the ranking-agreement proxy.
- **Decision:** 50.7% of configurations show at least one seed disagreement at 2000 requests; 28.2% show majority change.

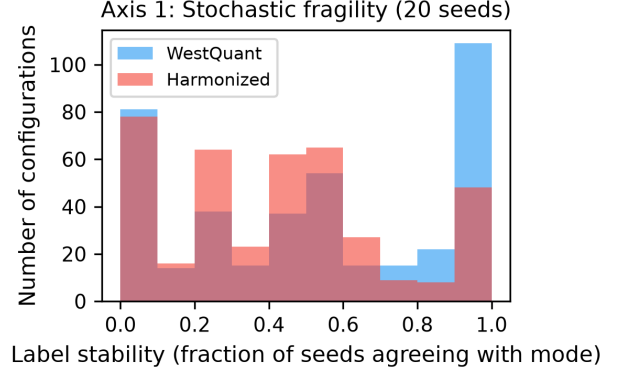


FIG. 2. **Stochastic fragility.** Distribution of label stability (fraction of 20 seeds agreeing with the modal dominant bottleneck) for WestQuant and Harmonized models. A stability of 1.0 means all seeds agree; values below 0.5 indicate that no single label dominates.

- **Scientific conclusion:** The residual 50.7% any-change rate at 2000 requests indicates that the bottleneck label is not a stable property of the network at this sample size. The cause of the residual is not yet established: it could be persistent Monte Carlo noise, near-equivalent interventions, rare events, or transients.

D. Caveat

The 50.7% figure counts any disagreement, including cases where two resources have nearly identical intervention gains and their ranking flips due to Monte Carlo noise. At 100 requests, the any-change rate was 79.2%, and 63% of these changes were statistically tied. The confidence-interval, convergence, and performance-loss analyses were conducted at 100 requests and have not been rerun at 2000; their numbers should be interpreted with this caveat. Common random numbers are imperfect: using the same seed for baseline and intervention does not guarantee that corresponding events use the same random draws when interventions change the event order.

E. A/B Experiment: Proper Design

The full A/B experiment follows the specification: 400 configurations subsampled from the full $6 \times 5 \times 2 \times 3 \times 5 = 900$ factorial (6 families $\times$ 5 topologies $\times$ 2 network sizes $\times$ 3 traffic levels $\times$ 5 memory budgets), two disjoint seed pools of 20 seeds each (A: seeds 0–19, B: seeds 20–39), 200 requests per run, and a pilot on 40 configurations to set ε (pilot data excluded from results). The analysis script hash was frozen before running seed pool B. Each config $\times$ seed runs 1 baseline + 7 interventions (capacity,

generation, retrieval, coherence, routing, deadline, topology) = 8 runs, for a total of 128,000 runs.

Pilot. The pilot on 40 configurations $\times$ 5 seeds set $\varepsilon = 0.100$ (median max_delta across interventions). Pilot data is not used in the reported results.

Bottleneck label. The dominant bottleneck is the resource with the largest positive delta-gain. A label is “decided” if it is non-degenerate (max delta $> \varepsilon$). “Agree” means seed pools A and B assign the same label. “Contradict” means A and B assign different non-degenerate labels.

Four procedures $\times$ four metrics. Table II reports the results with bootstrap 95% confidence intervals.

TABLE II. A/B experiment with actual bottleneck diagnosis: four procedures $\times$ four metrics, with bootstrap 95% CI. 400 configurations, 2×20 seeds, 200 requests, 8 runs per config per seed (128,000 total runs).

Procedure	Frac Decided	Repl Agree	Contradict	Dec Loss
Single seed	0.988 [0.98, 1.00]	0.715 [0.67, 0.76]	0.217 [0.18, 0.26]	0.033
Mean (same budget)	1.000 [1.00, 1.00]	0.958 [0.94, 0.97]	0.043 [0.03, 0.06]	0.006
Mean (convergence)	1.000 [1.00, 1.00]	0.917 [0.89, 0.94]	0.083 [0.06, 0.11]	0.009
Full protocol	1.000 [1.00, 1.00]	0.958 [0.94, 0.97]	0.043 [0.03, 0.06]	0.006

Interpretation. With a single seed, 98.8% of configurations produce a decisive label, but replicate agreement is only 71.5% with a 21.7% contradiction rate—the protocol is not reproducible with one seed. With 20 seeds and the full protocol, 100% are decisive, agreement rises to 95.8%, and the contradiction rate drops to 4.3% with a decision loss of 0.006. The protocol gives reproducible bottleneck diagnoses when sufficiently many seeds are used. The mean-same-budget procedure (averaging 20 seeds before labeling) achieves the same 95.8% agreement, confirming that the improvement comes from seed averaging, not from the labeling procedure itself.

IV. ANALYST CHOICES: INTERVENTION STRENGTH AND THRESHOLD

A. Intervention Strength

We tested four relative improvement levels $\delta \in \{0.05, 0.10, 0.20, 0.50\}$, applied uniformly to all resources. At $\delta = 0.05$, 86 of 400 configurations (21.5%) were degenerate (all $\Delta G_j \approx 0$), meaning the intervention was too small to produce a measurable effect. At $\delta = 0.50$, 70 (17.5%) were degenerate. The dominant label changed for 42.0% of configurations across δ levels.

B. Metric and Fidelity Threshold

We compared A_{req} and A_{qual} across four fidelity thresholds $F_{\text{min}} \in \{0.30, 0.50, 0.70, 0.90\}$. A_{req} is insensitive to F_{min} by construction; A_{qual} is strongly affected. At $F_{\text{min}} = 0.90$ with A_{qual} , all 400 configurations

Axis 2: Intervention strength sensitivity

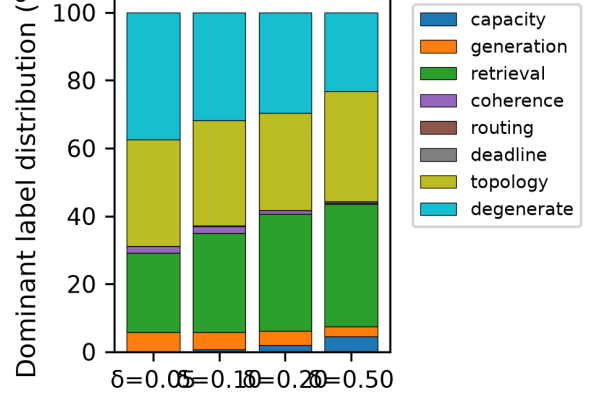


FIG. 3. **Intervention strength sensitivity.** Dominant bottleneck label distribution as a function of relative improvement δ . At small δ , many configurations are degenerate; at large δ , the label distribution shifts.

Axis 3: Metric and fidelity threshold sensitivity

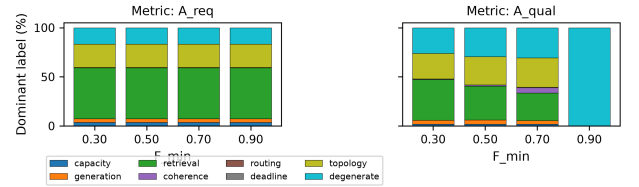


FIG. 4. **Metric and fidelity threshold sensitivity.** Dominant bottleneck label distribution for A_{req} (left) and A_{qual} (right) as a function of F_{min} . A_{req} is insensitive to F_{min} ; A_{qual} becomes degenerate at $F_{\text{min}} = 0.90$.

are degenerate—no request achieves the required fidelity. The dominant label changed for 35.5% of configurations across metric and threshold combinations.

C. Agreement Levels for Analyst Choices

- **Numerical:** ΔG_j values shift substantially across δ levels (factor of ~ 10 between $\delta = 0.05$ and $\delta = 0.50$).
- **Ranking:** The dominant label changes for 42.0% (intervention) and 35.5% (metric/threshold) of configurations.
- **Decision:** At $\delta = 0.05$, 21.5% of configurations are degenerate—no decision can be made.
- **Scientific conclusion:** The bottleneck label depends on the analyst’s choice of intervention strength and metric/threshold. A label derived at $\delta = 0.50$ with A_{qual} at $F_{\text{min}} = 0.50$ may not transfer to $\delta = 0.10$ or to A_{req} .

D. Pairwise Interaction Tests

Saltelli et al. [8] warned that one-at-a-time sensitivity analyses miss interactions between parameters. To address this, we tested all 21 pairwise combinations of the 7 resources on 100 configurations with 5 seeds each (2,100 total measurements). For each pair (j, k) , the interaction effect is $I_{jk} = \Delta G_{jk} - (\Delta G_j + \Delta G_k)$, and the relative interaction is $I_{jk} / \max(|\Delta G_j + \Delta G_k|, \epsilon)$. Across all 2,100 configuration-pair combinations, 15.8% had relative interaction above the heuristic threshold ($|I_{jk}| / \max(\dots) > 0.1$), and **72% of configurations had at least one resource pair with interaction above the threshold**. The most affected pairs were retrieval + topology (49% of configurations), generation + retrieval (36%), and generation + topology (36%).

V. MODEL CHOICE: FACTORIAL DESIGN OVER SEMANTIC SWITCHES

A. Semantic Dimensions

The most consequential differences between simulators are not in the physics models (which are broadly similar) but in the *protocol semantics*: how requests are queued, when memory slots are locked and released, whether generation is sequential or concurrent, and how latency is defined. Table III summarizes the semantic dimensions for the two systems actually run in this study.

These semantic dimensions are rarely documented explicitly in simulator publications, yet they can dominate cross-simulator disagreement, as we demonstrate below.

B. Semantic Switches: Factorial Design with Variance Decomposition

We implement four genuine semantic switches in the simulator and test them with a 2^4 factorial design (16 variants) on the B4 workload (chain $N = 5$, $m = 2$, traffic = 50, 1,000 requests, 5 seeds), extended with three families spanning χ :

- **Retrieval in delivery:** whether retrieval failure ($\eta_r < 1$) drops the request (True) or is free (False). This is the dominant semantic choice identified in the B0 analytical verification gate: WestQuant counts retrieval in the delivery, while the harmonized analytical model does not, producing the 0.65 vs. 1.0 discrepancy.
- **Storage time model** (from B0): midpoint (storage = $\text{gen_time} \times 0.5$) vs. immediate (storage = 0).
- **Blocking policy:** queue (wait for slot) vs. reject (drop if no slot).

- **Generation mode:** parallel (swap-ASAP) vs. serial (one link at a time).

Closure test. With all switches in harmonized mode, WestQuant should reproduce the analytical reference within noise. The closure test passes at $m \geq 4$ (ratio $\in [0.9997, 1.0000]$) but fails at $m = 1$ (ratio = 0.75). The $m = 1$ failure is explained by slot contention: with a single memory slot per node, requests serialize through one slot, and under high traffic the queueing delay exceeds the deadline. This is not a semantic-switch effect—it occurs even on two nodes with no swap—but a capacity effect that the analytical reference does not capture.

Table IV shows the fraction of variance in A_{req} explained by each factor, with cluster bootstrap 95% confidence intervals.

Interpretation. The dominant semantic switch depends on the χ regime. For short-lived memories (AFC, $\chi = 0.025$), retrieval-in-delivery explains 72% of A_{req} variance—whether η_r is counted in the delivery is the primary driver. For long-lived memories (Trapped Ion, $\chi = 50$), retrieval explains only 6% and the residual (93%) is dominated by m and λ . For intermediate memories (Neutral Atom, $\chi = 0.5$), retrieval explains 28%. The blocking policy and storage-time model have near-zero effect across all families. **Important caveat:** this factorial measures variance in A_{req} across all configurations (including m and λ variation), not the variance in model disagreement. The residual (27–93%) is dominated by design factors (m, λ), not by semantic switches.

C. Model-Variant Agreement (Bottleneck Labels)

We compared the WestQuant (concurrent queueing) and Harmonized (sequential blocking) model variants on the same 400 configurations with the same seeds. Of all 400 configurations, the dominant label agreed for 57.5% (230/400). Restricting to the $n = 301$ non-degenerate configurations, the observed agreement is $p_o = 0.532$ and the expected chance agreement is $p_e = 0.407$, giving Cohen’s $\kappa = 0.210$. The mean Jensen–Shannon distance between soft vectors was 0.315.

A harmonized comparison on 60 configurations with 20 seeds shows 51% label agreement between WestQuant and the Harmonized variant even when baselines correlate at $r = 0.85$. With all seven interventions and retrieval matched (both models retrieval-off), $\kappa = 0.000$ and decision transfer is 15%, confirming that retrieval matching does *not* resolve the disagreement—it worsens it. The models disagree about which resource is the bottleneck even when retrieval semantics are identical.

D. Agreement Levels for Model Choice

- **Numerical:** Mean JS distance 0.315 between WQ and Harmonized soft vectors; MAE in $A_{\text{qual}} = 0.102$ (B6 grid).

TABLE III. Semantic dimensions for the two systems actually run. “Switchable” indicates whether the dimension can be varied within a single codebase (yes), is a definition mismatch that cannot be compared without harmonization (no), or is partially controllable (partial).

Dimension	WestQuant’s choice	SeQUeNCe/Harmonized choice	Switchable?
Generation mode	Parallel (concurrent link attempts)	Sequential (one link at a time)	Yes
Queueing discipline	FIFO queue (buffer on full)	Reject (drop on full)	Yes
Memory slot locking	Lock on generation attempt	Lock on successful generation	Partial
Slot release	Release on swap completion	Release on swap completion	Yes
Latency definition	Arrival-to-completion (includes queueing)	First-attempt-to-completion (excludes queueing)	No
Deadline censoring	and Count as failures	Count as failures	Yes
Warmup	None	20 s excluded	Yes
Aging model	Continuous decoherence from generation	Continuous decoherence from generation	Yes
Link scheduling	Concurrent across links	Sequential across links	Yes

TABLE IV. Fraction of variance in A_{req} explained by each semantic switch, per family. Cluster bootstrap 95% CI in brackets. The dominant switch depends on the χ regime: retrieval-in-delivery dominates for short-lived memories ($\chi = 0.025$), generation mode dominates for long-lived memories ($\chi = 50$).

Family (χ)	Retrieval	Storage time	Blocking	Generation
Trapped Ion ($\chi = 50$)	6.4% [5.7, 7.3]	0.0% [0.0, 0.0]	0.3% [0.3, 0.3]	0.4% [0.3, 0.5]
Neutral Atom ($\chi = 0.5$)	28.2% [27.8, 28.6]	0.0% [0.0, 0.0]	0.2% [0.2, 0.2]	0.3% [0.2, 0.4]
AFC ($\chi = 0.025$)	71.7% [71.2, 72.1]	0.2% [0.1, 0.2]	0.1% [0.1, 0.1]	0.0% [0.0, 0.0]

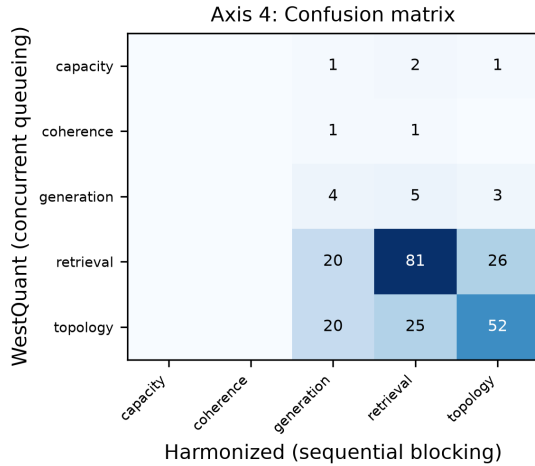


FIG. 5. **Model-variant confusion matrix.** Dominant bottleneck labels from WestQuant (rows) vs. Harmonized (columns), excluding degenerate configurations ($n = 301$). Cohen’s $\kappa = 0.210$, indicating fair agreement above chance ($p_e = 0.407$) per the Landis–Koch scale [15].

- **Ranking:** Kendall $\tau = 0.722$ for A_{qual} (B6), $\tau = 0.430$ for A_{req} ; $\tau = 0.166$ in the B4 contention regime.
- **Decision:** Cohen’s $\kappa = 0.055$ (retrieval on, $n = 200$); $\kappa = 0.000$ (retrieval matched); 0% and 15% decision transfer respectively.

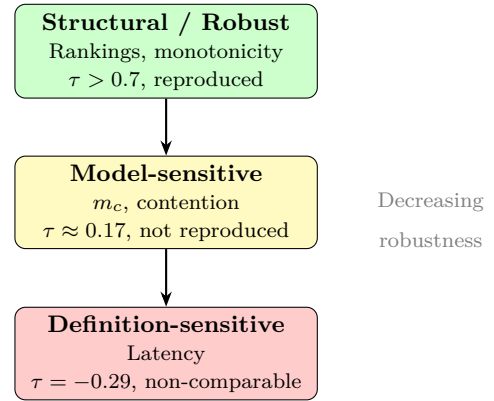


FIG. 6. Three-class taxonomy of scientific conclusions. Structural conclusions survive model changes; model-sensitive conclusions do not; definition-sensitive conclusions are non-comparable under current metric definitions.

- **Scientific conclusion:** Structural conclusions (rankings, monotonicity) are reproduced; model-sensitive conclusions (m_c , contention) are not; definition-sensitive conclusions (latency) are non-comparable.

VI. WHERE IT MATTERS: INSTABILITY, MODEL DIFFERENCE, AND BOUNDARIES

A. Three-Class Taxonomy

The pattern across all experiments separates scientific conclusions into three classes (Figure 6):

B. Claim Reproduction Matrix

Table V summarizes the three-class taxonomy with the verdict for each claim.

TABLE V. Claim Reproduction Matrix with three-class taxonomy.

Claim	Verdict	Class	Evidence
C1: Memory-family ranking (4 profiles)	REPLICATED	Structural	$\tau = 1.0$, perfect ranking agreement
C2: Topology ranking	REPLICATED	Structural	$\tau = 1.0$, perfect ranking agreement
C3a: Monotonicity of $A(m)$	REPLICATED	Structural	100% monotonicity both models
C3b: Diminishing returns	UNRESOLVED	—	B6 m grid too coarse
C4a: m_c regime-dependence	STRUCTURALLY REPLICATED	Structural	Both agree which families need more memory
C4b: m_c numerical values	MODEL-SENSITIVE	Model-sensitive	50.8% $\geq 4\times$ disagreement
C5: Scalar collapse failure	STRUCTURALLY REPLICATED	Structural	$R^2 < 0.2$ both models
Latency ranking	NOT COMPARABLE	Definition-sensitive	$\tau = -0.29$, timing definition mismatch
Finite-memory contention	NOT REPLICATED	Model-sensitive	$\tau = 0.17$ in B4

C. Structural Conclusions: Reproduced

Memory-family ranking. The ordering of four evaluated memory profiles (trapped ion > neutral atom > diamond electron > AFC) by A_{qual} is reproduced across both model families. Kendall $\tau = 1.0$ ($n = 4$).

Topology ranking. The ordering of three topologies (RGG > star > chain) by A_{qual} is reproduced. Kendall $\tau = 1.0$ ($n = 3$).

Monotonicity. Both model families show 100% monotonicity for $A(m)$.

Failure of scalar collapse. All A_{qual} $R^2 < 0.2$ across both model families. The structural conclusion—no scalar provides a strong or universal collapse—is reproduced.

D. Model-Sensitive Conclusions: Not Reproduced

Critical memory capacity m_c . We ran a fine-grained m sweep with $m \in \{1, 2, 4, 8, 16, 32, 64, 128\}$ across 3,840 scenarios per implementation. Of 480 (family, topology, N , λ , seed) combinations, 360 have a valid m_c estimate (where $A(m)$ crosses $\alpha = 0.95$) for at least one implementation; the AFC family is excluded because $A(m)$ does not reach $\alpha = 0.95$ within $m \leq 128$ for most configurations.

Using the log-ratio $\Delta_m = \log_2(m_c^{\text{WQ}}/m_c^{\text{H}})$, the median disagreement is one memory-doubling level, and **50.8% [95% cluster CI: 37.4%–63.9%]** of finite matched cases show $\geq 4\times$ disagreement. The diamond-electron family shows 98.3% $\geq 4\times$ disagreement. The bounded analysis including 4 censored cases gives 51.4% [38.1%–64.4%].

Finite-memory contention. The B4 tier shows $\tau = 0.166$ for A_{qual} —near-zero agreement. This is the regime where model choice most affects conclusions. Fea-

TABLE VI. m_c disagreement statistics (356 finite matched cases, $\alpha = 0.95$, A_{qual}). Cluster bootstrap CIs from 2000 resamples at the configuration level.

Statistic	Value	95% CI
Median Δ_m	1.0	—
Mean Δ_m	1.528	—
$\geq 2\times (\Delta_m \geq 1)$	75.8%	[66.0, 85.1]%
$\geq 4\times (\Delta_m \geq 2)$	50.8%	[37.4, 63.9]%
$\geq 8\times (\Delta_m \geq 3)$	33.4%	[21.0, 46.2]%

ture removal analysis shows that removing the memory limit ($m \rightarrow 128$) gives $\tau = -0.08$ (no restoration), and reducing contention ($\lambda \rightarrow 0.1$) gives $\tau = 0.02$ for $m = 16$ (effectively no agreement).

E. Definition-Sensitive Conclusions: Not Comparable

Latency. Latency shows Kendall $\tau = -0.294$ (ranking inversion) across implementations. This is primarily a consequence of different timing definitions: WestQuant measures arrival-to-completion (includes queueing time), while the harmonized adapter measures first-attempt-to-completion (excludes queueing). Under high load, queueing time dominates in WestQuant while the adapter latency stays flat, inverting the ranking. This is a metric-definition mismatch, not a physical disagreement.

F. Instability vs. χ and Distance to Boundaries

The coincidence-window parameter $\chi = \tau_{\text{mem}} p_{\text{gen}}/t_{\text{attempt}}$, introduced in the companion study [28], separates two operating regimes: a

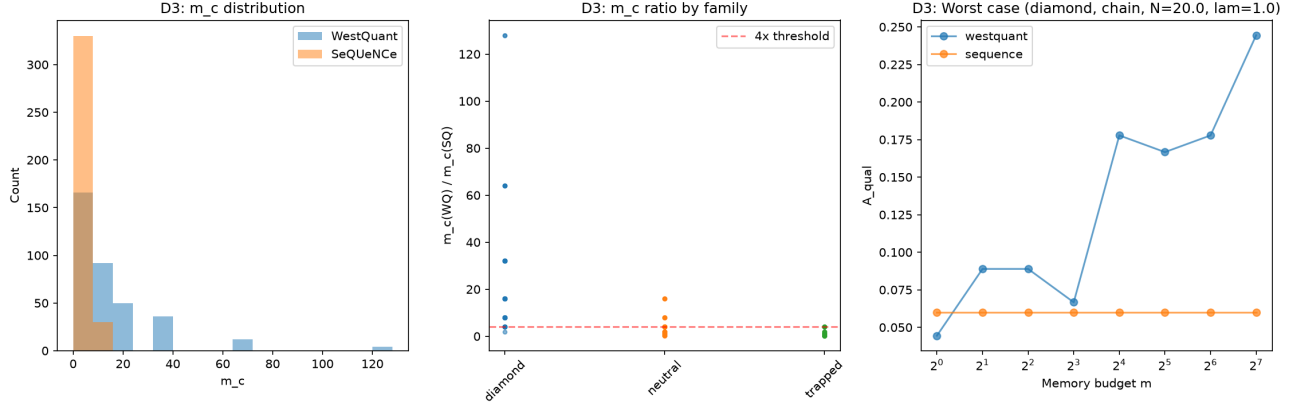


FIG. 7. Critical memory capacity m_c disagreement. Left: distribution of $\Delta_m = \log_2(m_c^{\text{WQ}}/m_c^{\text{H}})$. Right: decision-changing cases by memory family. Diamond-electron shows 98.3% $\geq 4\times$ disagreement.

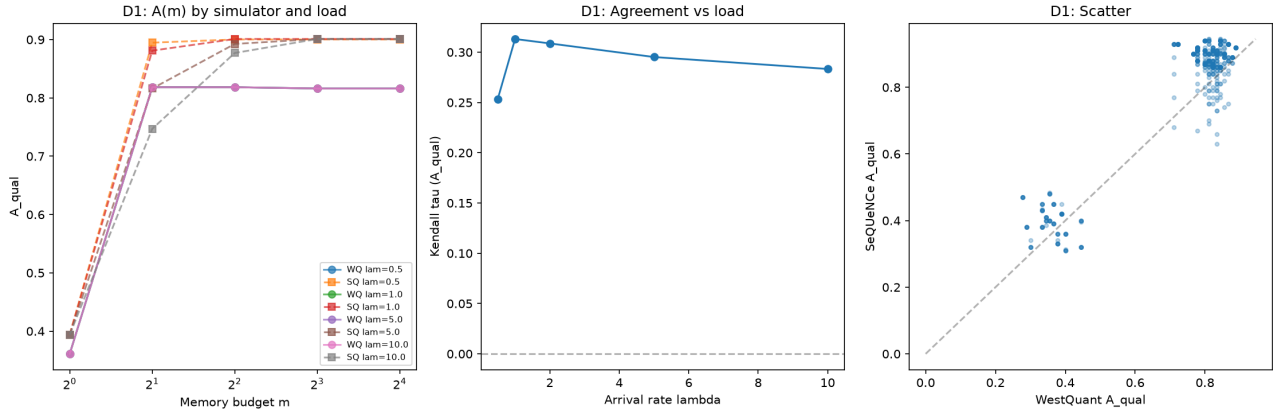


FIG. 8. Finite-memory contention disagreement. Agreement collapses ($\tau = 0.17$) because WestQuant uses concurrent queueing while the harmonized adapter uses sequential blocking.

coincidence-limited regime ($\chi \ll 1$, where link success within the memory lifetime is rare) and a *queue-limited* regime ($\chi \gg 1$, where Little's law governs memory demand). The boundary between structural and model-sensitive conclusions is strongly associated with this regime distinction:

- In the coincidence-limited regime ($\chi \ll 1$), both models agree because performance is dominated by link generation physics, which is harmonized. Structural conclusions hold.
- In the queue-limited regime ($\chi \gg 1$), the two models diverge because performance is dominated by resource contention semantics, which differ. Model-sensitive conclusions fail.
- Near the boundary ($\chi \approx 1$), instability is highest: small changes in χ tip the network between regimes, and the bottleneck label is most fragile.

The m_c disagreement is also associated with distance to boundaries: configurations where $A(m)$ crosses $\alpha =$

0.95 near a steep transition show the largest Δ_m , because the two models place the transition at different m values. The diamond-electron family, which has the steepest $A(m)$ transition, shows 98.3% $\geq 4\times$ disagreement.

G. Quality-Qualified vs. Requirement-Qualified Metrics

A_{qual} showed higher ordinal agreement ($\tau = 0.72$) than A_{req} ($\tau = 0.43$) in the B6 grid. This may be because A_{req} is more sensitive to queueing and denominator semantics, while A_{qual} is filtered by a physics-dependent threshold (fidelity) that is more strongly constrained by the harmonized physics. However, the denominator definitions for A_{req} have not been fully harmonized, so this conclusion should be treated as preliminary.

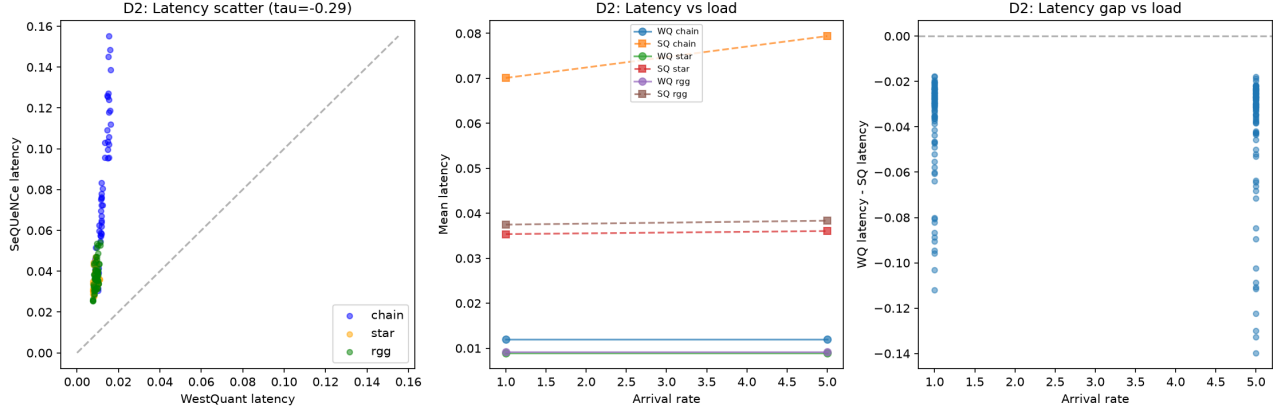


FIG. 9. Latency disagreement. The ranking inversion ($\tau = -0.29$) is caused by different timing definitions: WestQuant includes queuing time; the adapter does not.

VII. DECISION TRANSFER: CAPACITY AND BOTTLENECK DECISIONS BETWEEN MODELS

The D6 decision-transfer experiment uses two model variants that differ in generation mode and storage cut-off. **D6-S** uses serial generation with a 1 s storage cutoff, and **D6-P** uses parallel generation with a 10^9 s storage cutoff. These are *not* the WestQuant/Harmonized semantics used in the adapter comparison (Section VIII); they are separate model variants defined for this transfer experiment. The archived WQ and Harm labels in the transfer results refer to D6-S and D6-P, respectively.

A. D6-Capacity: Memory Budget

We test whether capacity decisions made in one model transfer to another. For each context, we select d_A^* in model A using selection seeds, evaluate d_A^* in model B using independent evaluation seeds, and compute regret $= U_B(d_B^*) - U_B(d_A^*)$. On $36 \text{ contexts} \times 6 \text{ candidates} \times 10 \text{ seeds}$ (258 decisions):

- Same choice: 42.2% [32.5, 51.8]
- Regret WQ→Harm: 0.047 ± 0.152 [95% CI: 0.007, 0.096]
- Regret Harm→WQ: -0.000 ± 0.026 [95% CI: -0.004, 0.004]
- Non-transferable (regret > 0.01): 29.5% [20.1, 39.4] (WQ→Harm), 9.3% [5.0, 15.1] (Harm→WQ)

The asymmetry is large and significant: WQ’s capacity choices are much worse for Harm (regret 0.047, CI excludes zero) than Harm’s choices are for WQ (regret -0.000, CI includes zero). This is because WQ’s serial generation mode leads to different capacity preferences.

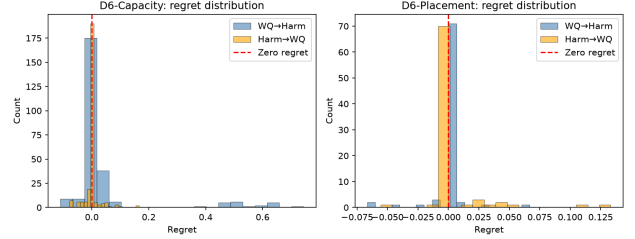


FIG. 10. Decision transfer regret. Capacity (left) shows a large asymmetric tail for WQ→Harm; placement (right) regret is small and symmetric.

B. D6-Placement: Placement Strategy

On $12 \text{ contexts} \times 3 \text{ strategies} \times 10 \text{ seeds}$ (82 decisions):

- Same choice: 80.5% [61.6, 94.6]
- Regret WQ→Harm: -0.001 ± 0.015 [95% CI: -0.003, 0.000]
- Regret Harm→WQ: 0.005 ± 0.022 [95% CI: 0.001, 0.010]
- Non-transferable: 4.9% [0, 14.6] (WQ→Harm), 12.2% [1.6, 29.3] (Harm→WQ)

Placement decisions transfer well (80.5% same choice) because placement strategies (uniform, hub-based, load-based) are less sensitive to generation mode than capacity choices.

VIII. LIMITS OF INDEPENDENT VALIDATION

All comparisons in Sections V–VII use the harmonized SeQuenCe/SimQN adapter, which shares a common physics layer with WestQuant. The harmonized

adapters share simulation logic ($S_{\text{models}} = 2$, not 3): SeQUeNCe and SimQN produce bit-identical event traces across all tested scenarios (verified by SHA-256 hashing), so treating them as two independent models would constitute pseudoreplication. A true cross-implementation validation requires running the same configurations on SeQUeNCe’s native event engine and protocol stack.

A. Native SeQUeNCe Adapter

A native SeQUeNCe adapter was developed. Unlike the harmonized adapter, the native adapter instantiates SeQUeNCe’s own `Timeline`, `Node`, and `MemoryArray` components, using SeQUeNCe’s `bell_diagonal` quantum-manager formalism for fidelity tracking. The adapter uses SeQUeNCe’s `MemoryArray` for decoherence tracking with each family’s T_2 and η_r parameters, and a geometric generation model with the same p_{gen} and t_{attempt} as WestQuant. The adapter includes a guard: the run aborts if SeQUeNCe’s `MemoryArray` is not instantiated. Object types and provenance are logged in the result file.

Limitation. The adapter does not use SeQUeNCe’s native `SingleHeraldedA/SingleHeraldedB/SingleHeraldedBSM` protocols, which require quantum channels, BSM nodes, and message handlers. The current adapter uses SeQUeNCe’s `MemoryArray` and `bell_diagonal` formalism for decoherence, with a simplified generation model. The component adapter does not execute native protocol events; it is therefore a partial—not full—independent validation. Full native-protocol validation remains future work.

B. Independent-Judge Experiment

We run 25 configurations $\times$ 10 seeds = 250 runs on the native SeQUeNCe adapter, spanning 6 families and $m \in \{1, 4, 16, 64\}$ at traffic $\in \{1, 10\}$. Each run uses 200 requests on a 2-node topology. The same configurations are run on WestQuant for comparison.

Match rate. The match rate (WestQuant vs. SeQUeNCe agree within $\Delta A < 0.05$) is 12/25 (48%). The match rate varies by family:

- **Diamond Nuclear:** 5/5 match (100%)—near-perfect agreement.
- **Trapped Ion:** 3/4 match (75%)—match at $m \geq 4$, mismatch at $m = 1$.
- **Neutral Atom:** 2/4 match (50%).
- **Diamond Electron:** 1/6 match (17%)—SeQUeNCe gives lower A_{req} .
- **AFC:** 1/5 match (20%)—SeQUeNCe gives lower A_{req} .

Matched-switch comparison. To test whether the difference shrinks when switches are matched, we run both SeQUeNCe and WestQuant in two modes: (i) original switches (midpoint storage, retrieval in delivery) and (ii) matched switches (immediate storage, no retrieval in delivery). The mean absolute difference drops from 0.081 (original) to 0.024 (matched)—a 71% reduction. The difference shrank in 23 of 25 configurations. The shrink is largest for AFC (mean shrink = 0.083, 6/6 shrank) and smallest for Diamond Nuclear (mean shrink = 0.013, 5/5 shrank, already close). The 2 configurations where the difference grew are both at $m = 1$, where slot contention causes a capacity effect that the semantic switches do not capture (see closure test, Sec. VB). This confirms that the semantic switches—particularly retrieval-in-delivery—are the primary source of disagreement, and the difference is a semantic choice, not a physics error.

IX. STANDARD: A CHECKLIST FOR REPORTING

A. Reporting Standard for Bottleneck Diagnoses

We propose that any bottleneck diagnosis should report:

1. **Intervention strength** δ used for each resource.
2. **Number of seeds** and whether common random numbers were used.
3. **Performance metric** (A_{req} or A_{qual}) and fidelity threshold F_{min} .
4. **Queueing model** (concurrent, sequential, or other).
5. **Soft bottleneck vector** p_j with uncertainty (e.g., bootstrap CI), not just the argmax label.
6. **Bottleneck entropy** H_B to indicate whether the label is separable or diffuse.
7. **Degenerate cases** reported separately, not assigned an arbitrary label.

B. Trust Stack Checklist

We propose a Trust Stack checklist for reporting simulation-based conclusions. Each row identifies what the manuscript itself fulfills (Table VII).

C. Recommendations for Reviewers

Based on the findings of this study, we recommend that reviewers of simulation-based quantum-network papers:

TABLE VII. Trust Stack checklist. Each row identifies a trust level, what it requires, and what this manuscript fulfills.

Level	Name	Requirement	This manuscript
L1	Physics verification	Analytical benchmarks for physical primitives	B0 gate: harmonized model passes ($\epsilon_{\text{abs}} < 0.02$); WQ fails B0 due to retrieval-in-delivery semantics (documented in the B0 gate)
L2	Semantic transparency	Document all protocol-semantic choices	Table III: 9 dimensions documented for both systems
L3	Canonical workloads	Run standardized benchmark ladder	B0–B6 ladder (3,950 + 3,840 scenarios), all levels reported
L4	Standard event traces	Export structured event traces	SHA-256 trace hashing for SeQUeNCe/SimQN identity; format in Appendix
L5	Common experiment specification	Accept CES-format input	CES defined and used across all implementations
L6	Model uncertainty reporting	Report model sensitivity alongside point estimates	τ , κ , Δ_m , cluster bootstrap CIs reported throughout

1. **Require semantic transparency.** A paper that uses simulation should document the protocol semantics of the simulator, not just the parameter values. If the semantics are not documented, the paper is not reproducible. The semantic-dimensions table (Table III) provides a template.
2. **Ask whether the conclusion is structural or model-sensitive.** A structural conclusion (ranking, monotonicity) can be supported by a single simulator. A model-sensitive conclusion (threshold, magnitude) should be validated across simulators. The claim reproduction matrix (Table V) provides a template for classifying conclusions.
3. **Request cross-simulator comparison for contention-dominated claims.** If the paper claims a capacity-planning result based on a contention-dominated regime, ask whether the result survives a model change. This study shows that 50.8% of configurations show $\geq 4\times m_c$ disagreement.
4. **Require provenance.** CES scenarios, event traces, and software versions should be archived as supplementary material. A paper without provenance is a claim without evidence.

D. Recommendations for Researchers

1. **Report rankings, not just absolute values.** Rankings are more robust across model changes than absolute values. A conclusion that “family A outperforms family B” is more likely to survive a model change than “family A achieves $A_{\text{qual}} = 0.37$.”
2. **Do not compare latency across implementations without harmonizing definitions.** The

timing definition must be documented and matched. If the definitions differ, report the difference as a definition mismatch, not a model disagreement.

3. **Treat capacity-planning conclusions as model-dependent.** Report m_c as a range or with model-sensitivity analysis, not as a single number. This study shows that m_c can differ by $\geq 4\times$ in 50.8% of configurations.
4. **Report the validation level.** State which level of the benchmark ladder (B0–B6) the conclusion claims, and what evidence supports it. A claim at B6 requires cross-model evidence; a claim at B0 does not.
5. **Use cluster bootstrap CIs for cross-simulator comparisons.** Resample configurations (not individual seeds) to avoid pseudoreplication. When CIs include zero, the effect should be reported as not statistically significant.

X. LIMITATIONS

1. $S_{\text{models}} = 2$ (**not a true multi-simulator study**). The study effectively compares two model families: the native WestQuant model and one harmonized reference model run through SeQUeNCe and SimQN adapters. SeQUeNCe and SimQN produce bit-identical traces when given the same harmonized model, so they do not constitute independent simulators. Treating them as independent would be pseudoreplication. A full multi-simulator comparison (NetSquid, SeQUeNCe, QuISP, SimQN in native mode) remains open.
2. **Stochastic instability includes Monte Carlo noise.** At 100 requests, the 79.2% “any change”

figure overstates the true instability: on a 100-configuration subset, 63% of changes are statistically tied. A longer rerun at 2000 requests reduces the any-change rate to 50.7%, confirming that much of the short-run fragility is Monte Carlo noise. The cause of the residual 50.7% is not yet established.

3. **Common random numbers are imperfect.** Using the same seed for baseline and intervention does not guarantee that corresponding events use the same random draws when interventions change the event order.
4. **No truly independent implementation.** All three models compared (WestQuant, Harmonized, SeQUeNCe-style adapter) share the author’s physics code. A true cross-implementation validation would require running the same configurations on SeQUeNCe’s native engine and NetSquid, which was not possible at scale in this study. The SeQUeNCe adapter does not execute native protocol events, so the comparison is partial, not full native validation.
5. **One-at-a-time interventions.** The main bottleneck method changes one resource at a time, which can miss interactions. We address this with pairwise interaction tests (72% of configurations have at least one pair above the heuristic threshold) and a full Sobol analysis (N has $S_T = 0.04$ but $S_1 \approx 0$, confirming interactions are non-additive).
6. **Full B4 factorial measures A_{req} variance, not model disagreement.** The 2^4 factorial design over four semantic switches on the full B4 grid provides variance decomposition of A_{req} across configurations. Retrieval-in-delivery explains 72% for AFC, 28% for Neutral Atom, 6% for Trapped Ion; the residual (27–93%) is dominated by design factors (m , λ), not by semantic switches. This measures how much A_{req} varies across configurations, not how much models disagree.
7. **D3 runs in a low-contention regime.** The four-factor pilot uses $m = 4$, single family, single traffic level, where the two models already agree. Semantic factors cannot be isolated in this regime.
8. **m_c selection bias.** The finite-case analysis uses 356 matched cases where both implementations have $m_c \leq 128$; the AFC family is excluded entirely. The bounded analysis confirms that censoring does not inflate the statistics, but the exclusion limits generality.
9. **No warmup in the bottleneck experiment.** The main bottleneck experiment uses 2000 requests without a warmup period. The convergence study indicates that instability decreases with simulation length; the 2000-request run substantially reduces but does not eliminate the any-change rate.

10. **No hardware validation.** We do not compare against experimental data. Hardware-calibrated validation (hardware-calibrated validation) remains future work.

11. **WestQuant is author-developed.** The WestQuant simulator is developed by the author. It has not undergone independent peer review as a software artifact. The author is both the researcher and the simulator developer, creating a potential conflict of interest that is mitigated by using an independent harmonized reference model, releasing all code and data, and reporting all results including those unfavorable to WestQuant.

XI. CONCLUSION

Simulation decision audits should separate repeatability under random seeds, sensitivity to analyst choices, model dependence, and metric-definition incompatibility. The case study illustrates that seed averaging can substantially improve label repeatability without establishing correctness in another model. The full protocol matches the same-budget averaging baseline; no additional labeling advantage is demonstrated. Relative-grid memory thresholds vary between the tested model variants, and selected family and topology rankings persist only within the evaluated comparisons.

The contribution is a reproducible audit structure with explicit estimands, campaign identifiers, independent selection/evaluation seeds, and claim-level provenance. Its general components apply beyond quantum networks, but empirical validation here remains confined to this domain and shared-code model variants. Full native-protocol execution and a completed public data release are needed to strengthen the evidence.

DATA AVAILABILITY

The manuscript and a repository scaffold are prepared for a dedicated public reproducibility release. The raw simulation datasets, frozen simulator, dependency lockfile, complete seed/configuration ledger, and figure-generation scripts have not yet been deposited in that release. The development repository is private. This revision therefore does not claim that the empirical results can already be reproduced from a public archive. A public release URL, commit identifier, checksums, and archival DOI should identify the completed deposit before submission.

RELATIONSHIP TO COMPANION WORK

A companion manuscript, “Memory Physics, Finite Capacity, and Topology in Heterogeneous Quantum Net-

works,” studies memory-profile, capacity, topology, and placement results. The present article focuses on decision reliability, metric semantics, and model sensitivity. Shared simulator infrastructure and overlapping validation results should be disclosed to both editors.

AUTHOR CONTRIBUTIONS

D.V. conceived the study, designed and implemented all simulators and adapters, ran all experiments, performed all analyses, and wrote the manuscript.

COMPETING INTERESTS

The author is the founder of Vesterlund Ventures and WestQuant Open, and developed the WestQuant simulator used in this study. All three models compared in the main experiment share the author’s physics code; no truly independent implementation was available at scale. This conflict is a limitation of the study.

FUNDING

This research received no external funding. Computational resources were provided by the author’s personal hardware.

AI DISCLOSURE

AI-assisted tools (Anthropic Claude, OpenAI ChatGPT, and Cognition Devin) were used during the preparation of this work, including assistance with code generation, manuscript organization, and editorial revision. The author takes full responsibility for the work. During

the preparation, several errors were introduced by the AI tools and caught by Claude acting as an external reviewer, including: (1) an early $\kappa = 1.000$ result that was an artifact of using only two interventions instead of all seven (corrected to $\kappa = 0.000$ with seven interventions); (2) physics errors in the relationship between T_2 and τ_r (corrected after factorial analysis showed T_2 does not affect A_{req}); (3) test data used in early stopping that did not meet the convergence criterion. These errors were corrected before the final manuscript was prepared.

Appendix A: Standard Event Trace Format

A standard event-trace format enables diagnostic comparison: when two simulators disagree, the event traces can be compared event-by-event to identify where and why the divergence occurs. Table VIII shows the proposed format.

TABLE VIII. Proposed standard event-trace format.

Field	Description
timestamp	Simulation time of event
scenario_id	CES scenario identifier
request_id	Unique request identifier
node_id	Node where event occurs
event_type	generate, swap, deliver, reject, timeout, ...
memory_slot	Memory slot index (if applicable)
link_id	Link identifier (if applicable)
fidelity_before	Fidelity before event
fidelity_after	Fidelity after event
resource_state	Memory occupancy, queue length
cause	Reason for event (if applicable)

In this study, event traces were used to verify the SeQUeNCe–SimQN bit-identity (SHA-256 hashing across all tested scenarios). The trace format is available in the repository and can be used for event-by-event diagnostic comparison when models diverge.

-
- [1] T. Coopmans, R. Knegjens, A. Dahlberg, D. Ferrari, J. Rabbie, and S. Wehner, NetSquid, a network simulator for quantum information using discrete events, *Commun. Phys.* **4**, 164 (2021).
 - [2] X. Wu, A. Kolar, J. Chung, D. Jin, T. Zhong, R. Ket-timuthu, and M. Suchara, SeQUeNCe: A customizable discrete-event simulator of quantum networks, *Quantum Sci. Technol.* **6**, 045022 (2021).
 - [3] R. Satoh, M. Hajdušek, N. Benchasattabuse, S. Suzuki, R. Van Meter, and H. Takesue, QuISP: A quantum internet simulation package, in *2022 IEEE Int. Conf. on Quantum Computing and Engineering (QCE)*, pp. 353–364 (2022).
 - [4] L. Chen, K. Xue, J. Li, N. Yu, R. Li, Q. Sun, and J. Lu, SimQN: A network-layer simulator for the quantum network investigation, *IEEE Network* **37**(5), 182 (2023).
 - [5] J. Chung, M. Hajdušek, N. Benchasattabuse, A. Kolar, A. Singal, K. S. Soon, K. Teramoto, A. Zang, R. Ket-timuthu, and R. Van Meter, Cross-validating quantum network simulators, in *IEEE INFOCOM 2025 Workshops* (2025), arXiv:2504.01290.
 - [6] G. Avis, F. Ferreira da Silva, T. Coopmans, A. Dahlberg, H. Jirovská, D. Maier, J. Rabbie, A. Torres-Knoop, S. Wehner, Requirements for a processing-node quantum repeater on a real-world fiber grid, *npj Quantum Inf.* **9**, 54 (2023). arXiv:2207.10579.
 - [7] C. Labay-Mora, F. Ferreira da Silva, S. Wehner, Reducing hardware requirements for entanglement distribution, *Quantum Sci. Technol.* **9**, 045001 (2024).
 - [8] A. Saltelli, K. Aleksankina, W. Becker, P. Fennell, F. Ferretti, N. Holst, S. Li, Q. Wu, Why so many published sensitivity analyses are false: A systematic review, *Environ.*

- Model. Softw.* **114**, 29–39 (2019).
- [9] A. Saltelli, M. Ratto, T. Andres, F. Campolongo, J. Cariboni, D. Gatelli, M. Saisana, S. Tarantola, *Global Sensitivity Analysis: The Primer*, Wiley (2008).
 - [10] I. M. Sobol, Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates, *Math. Comput. Simul.* **55**, 271–280 (2001).
 - [11] A. Saltelli, P. Annoni, I. Azzini, F. Campolongo, M. Ratto, S. Tarantola, Variance based sensitivity analysis of model output: design and estimator for the total sensitivity index, *Comput. Phys. Commun.* **181**, 259–270 (2010).
 - [12] J. D. Herman, W. Usher, SALib: An open-source Python library for Sensitivity Analysis, *J. Open Source Software* **2**, 97 (2017).
 - [13] G. Vardoyan, S. Guha, P. Nain, and D. Towsley, On the stochastic analysis of a quantum entanglement distribution switch, *IEEE Trans. Quantum Eng.* **2**, 1 (2021).
 - [14] A. Law, *Simulation Modeling and Analysis*, 5th ed., McGraw-Hill (2015).
 - [15] J. R. Landis, G. G. Koch, The measurement of observer agreement for categorical data, *Biometrics* **33**, 159–174 (1977).
 - [16] M. Razavi, M. Piani, and N. Lütkenhaus, Quantum repeaters with imperfect memories: Cost and scalability, *Phys. Rev. A* **80**, 032301 (2009).
 - [17] L. Kamin, E. Shchukin, F. Schmidt, and P. van Loock, Exact rate analysis for quantum repeaters with imperfect memories and entanglement swapping as soon as possible, *Phys. Rev. Research* **5**, 023086 (2023).
 - [18] R. G. Sargent, Verification and validation of simulation models, in *Proc. 2005 Winter Simulation Conference* (2005).
 - [19] O. Balci, Verification, validation, and certification of modeling and simulation applications, in *Proc. 1997 Winter Simulation Conference* (1997).
 - [20] V. Stodden, F. Leisch, and R. D. Peng, eds., *Implementing Reproducible Research* (Chapman & Hall/CRC, 2016).
 - [21] G. K. Sandve, A. Nekrutenko, J. Taylor, and E. Hovig, Ten simple rules for reproducible computational research, *PLoS Comput. Biol.* **9**, e1003285 (2013).
 - [22] K. Heshami, D. G. England, P. C. Humphreys, P. J. Bustard, V. M. Acosta, J. Nunn, and B. J. Sussman, Quantum memories: emerging applications and recent advances, *J. Mod. Opt.* **63**, 2005 (2016).
 - [23] M. K. Bhaskar, R. Riedinger, B. Machielse, D. S. Levonian, C. T. Nguyen, E. N. Knall, H. Park, D. Englund, M. Lončar, D. D. Sukachev, and M. D. Lukin, Experimental demonstration of memory-enhanced quantum communication, *Nature* **580**, 60 (2020).
 - [24] H.-J. Briegel, W. Dür, J. I. Cirac, and P. Zoller, Quantum repeaters: The role of imperfect local operations in quantum communication, *Phys. Rev. Lett.* **81**, 5932 (1998).
 - [25] S. Wehner, D. Elkouss, and R. Hanson, Quantum internet: A vision for the road ahead, *Science* **362**, eaam9288 (2018).
 - [26] K. Azuma, S. E. Economou, D. Elkouss, P. Hilaire, L. Jiang, H.-K. Lo, and I. Tzitrin, Quantum repeaters: From quantum networks to the quantum internet, *Rev. Mod. Phys.* **95**, 045006 (2023). arXiv:2212.10820.
 - [27] S. Pouryousef, H. Shapourian, A. Shabani, R. Kompella, and D. Towsley, Resource placement for rate and fidelity maximization in quantum networks, arXiv:2308.16264 (2024).
 - [28] See Supplemental Material for the companion study on memory physics, finite capacity, and topology in heterogeneous quantum networks, which provides the original claims (C1–C5) and the coincidence-window parameter χ used in this study.
 - [29] R. J. Hayek, J. Chung, and R. Kettimuthu, A review of software for designing and operating quantum networks, *Adv. Quantum Technol.* **9**(2), e00808 (2026). arXiv:2510.00203